

Towards a taxonomy of group pattern discovery algorithms from spatio-temporal trajectories

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I would like to dedicate this dissertation to my loving family and especially to my little boss

xxx . . .

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work, except as specified in the text.

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Abstract

Over the years, the increasing development of location acquisition devices have generated a significant amount of spatio-temporal data. This data can be further analysed in search for some interesting patterns, new information, or to construct predictive models such as next location prediction. While data mining has been practiced for many years, the spatio-temporal data mining is a relatively new discipline and therefore requires certain standardization of the processes involved. The goal of this dissertation is to contribute to the future research and development of group pattern discovery algorithms from spatio-temporal data by providing a conceptual framework for group pattern mining, followed by a comprehensive classification of existing models. The proposed framework includes both regular and big data as well as data stream processing models which to the best of author's knowledge is the first attempt of presenting them in this context and including them in the data mining process. Furthermore, there is not any detailed publication on the classification of group pattern discovery algorithms from spatio-temporal trajectories which includes the state-of-the-art algorithms in this research area. The proposed dissertation will take a form of a survey/taxonomy of the existing group discovery models from spatio-temporal trajectories. Additionally, since this paper is considered as the step towards developing a novel algorithm in the future, the outcome of this classification results in a concept of an "ideal" model for group discovery from spatio-temporal trajectories that addresses current gaps in this research area. Namely, the lack of a distributed model capable of handling data streams which is also parameter-light and can have multiple threshold options to efficiently discover various groups depending on the requirements.

Table of contents

List of figures	xiii
List of tables	xv
1 Introduction	1
2 Related work	3
2.1 Statement of Problem	7
2.2 Goals and Objectives	8
3 Methods	9
3.1 Methodology	9
3.2 Group and movement parameters	10
3.3 Framework	12
4 Classification	15
4.1 Definitions and Key Concepts	15
4.2 Spatio-temporal Input Data	21
4.2.1 Data Attributes	21
4.2.2 Data Type Source	22
4.2.3 Data Size	23
4.3 Pre-processing	24
4.4 Thresholds	26
4.5 Taxonomy	29
5 Algorithm Design	31
5.1 Static data algorithms	32
5.2 Streaming data algorithms	36
5.3 Distributed/parallel algorithms	39

6	Universal Distributed and Parameter-free (USDP) algorithm	41
7	Conclusion	45
	References	47

List of figures

3.1	Framework	13
4.1	Moving Point Objects	17
4.2	Spatio-Temporal Data Sample	22
4.3	Thresholds Differences 1	27
4.4	Thresholds Differences 2	28
4.5	Thresholds Differences 3	28

List of tables

2.1	Commonalities and differences between selected group patterns	4
3.1	Classification - Group Patterns	11
4.1	Table of Notations	16
4.2	Taxonomy - Group Patterns	30

Chapter 1

Introduction

The growth of tracking technologies and development of location acquisition devices have generated big volumes of spatio-temporal data. This vast amount of data can be further analysed in order to bring valuable outcome by either discovering interesting findings and patterns, or by creating predictive models based on the given input. While data mining has been practiced for many years, the spatio-temporal data mining is a relatively new discipline and requires different approaches and techniques due to the nature of the input data. Spatio-temporal data is different from relational data as it adds both spatial and temporal attributes to regular observations. Given the example of a web search, in addition to the search request and related information, the spatio-temporal nature of the data would mean the stored request also comes with time and location details from where the search was made. Similarly, a CCTV, or other monitoring device would record the motion, or image of the given area together with its geo-coordinates and time the video, or image was taken. Shekhar et al. differentiates three distinct types of data attributes in the context of spatio-temporal data (Shekhar et al., 2015). Aside from the spatial and temporal attributes, there are also non-spatiotemporal ones which provide additional characteristics of objects which could be for example their name (nominal), or exchange rate (ratio) among others. Spatial attributes can either define a location by providing for instance its geo-coordinates, spatial size, area or shape while temporal attributes can be anything from timestamps, snapshots, or duration. One of the challenges in spatio-temporal data mining process is dealing with spatio-temporal relationships. One of the approaches could be to transform them into a traditional data input so they can be modeled using traditional data mining techniques. Another - to find new ways of incorporating both spatial and temporal attributes into the data mining process.

In addition to the complexity of the spatio-temporal data itself, scalability comes as another challenge. With endless data generated every second heading towards zettabytes in the nearest future, the data mining algorithms need to cope with the volumes. Seeing the

temporal aspect of such data and the increasing development of streaming technologies, the data mining process requires new solutions to speed up the processing. This could be for instance new data structures to store the relationship rather than the information which is too heavy to be processed (Tang et al., 2013), or distributed/parallel processing models (Orakzai et al., 2016; Vatsavai et al., 2012).

Spatio-temporal data types can be further classified basing on the way how space and time are used for data collection and representation. Atluri et al. specifies four categories of spatio-temporal data types (Atluri et al., 2017). First of them is event data which includes discrete events occurring at point locations and times which can find its real-world application in crime detection. Another is raster data which can be best represented in the example for fMRI scans of brain activity. Point reference data is the third category with its application in surface temperature collection and lastly the trajectory data. Trajectory data is focused on measuring the trajectories of moving objects.

In his survey on trajectory data mining Zheng classified four types of spatio-temporal data based on the moving objects (Zheng, 2015) These types include trajectories of people, vehicles, animals and natural phenomena and each of them assuming various approaches of data mining. These trajectories have various applications including in intelligent transportation systems, social networks, or in commercial use (for e.g. to offer products based on the next location of the user). Given the example of transportation many taxis are equipped with GPS sensors to report on their locations to the transportation system or applications for the users to find their next available taxi. Scientists record trajectories of animals to analyse their behaviours and patterns for further research.

Trajectory data mining has become an interdisciplinary field attracting specialists from many areas including computer science but also sociology, biology, or geography. Further in his survey on trajectories, Zheng identifies four major categories of patterns from either single or group trajectories (Zheng, 2015). These include trajectory clustering, sequential patterns, periodic patterns and moving together patterns. The first one aims at grouping the similar trajectories into patterns. The second looks at common sequences of locations in similar time intervals. The third one as the name suggests analyses the periodic occurrence of patterns. Lastly, the moving together patterns or in other words the group pattern mining from spatio-temporal trajectories will be the research area for this dissertation which is further discussed in the following sections.

Chapter 2

Related work

There have been several distinct approaches to group pattern mining from spatio-temporal trajectories. One of the widely discussed approaches is flock pattern which is also considered one of the earliest group pattern discovery approaches (Benkert et al., 2008; Gudmundsson, 2004; Laube and Imfeld, 2002). The second approach, convoy, argues that the first model is not sufficient for correct group discovery due to its limitation of pre-defined group search region which leads to group omissions (also known as so-called lossy-flock problem). Instead, this second approach proposes to capture trajectory pattern of any shape by applying the density-based clustering (Jeung et al., 2008; Zheng, 2015). The third approach, swarm identifies another challenge with both aforementioned approaches as they have a strict requirement on the k -consecutive time points for group discovery (Li et al., 2010). This may result in ignoring certain groups depending on the k -consecutive time set up. For instance if k is set up to 9 but the objects would travel together during $k=7$, then this group would have been missed. Instead, the swarm approach introduces a cluster of objects that can last for at least k but non-consecutive time stamps. While all these approaches use static datasets and additionally convoy and swarm require loading the entire trajectories into memories in order to identify the clusters, the traveling companion approach implements a data structure called *traveling buddy* (Tang et al., 2012) which allows finding patterns from streamed data and therefore sets up a ground for further approaches. Lastly, the most recent approach - loose traveling companion pattern addresses certain limitations of the previous approaches by focusing on human and not object/animal trajectories as presented in the previous patterns and considering indoor environment. A key observation is made that human trajectories are different to these of objects and animals as people can belong to one group although they have different trajectory paths (which is different to convoy) (Naserian et al., 2018).

The aforementioned approaches do not exhaust the group discovery patterns in spatio-temporal trajectories. Gathering, group pattern, moving cluster, evolving cluster, or weakly

Table 2.1 Commonalities and differences between selected group patterns

pattern	duration threshold	group formation	group membership	input data
flock	k-consecutive	pre-defined	fixed	static
convoy	k-consecutive	any	fixed	static
swarm	non-consecutive	any	fixed	static
gathering	non-consecutive	any	evolving	static
group pattern	non-consecutive	pre-defined	fixed	static
traveling companion	k-consecutive	any	fixed	streaming
loose companion	non-consecutive	any	dynamic	streaming
moving cluster	k-consecutive	any/cluster	dynamic	static
evolving convoy	k-consecutive	any/cluster	dynamic	static
WCGMP¹	non-consecutive	any/cluster	dynamic	static
LTCP²	k-consecutive	any/cluster	dynamic	streaming
WCLTP³	non-consecutive	any/cluster	dynamic	streaming

1 - Weakly Consistent Group Movement Pattern, 2 - Loose Traveling Companion Pattern, 3 - Weakly Continuous Loose Traveling Pattern

consistent group movement pattern are just few other examples that have contributed to the subject area. Furthermore, each of the classical approaches has its modifications such as variable flock which permits the members of the group to change during the time-span. Further analysis of these algorithms reveals certain commonalities they have as well as distinct features. Some of them have been presented in Table 2.1

Based on the reviewed literature the main features can be grouped around the continuity of the group (consecutive/non-consecutive time), how the group is formed (group formation), whether the members can change during the lifespan (group membership) and what data is used for pattern mining (static/streaming data). Additionally, each of the approaches can be seen from the scalability point of view and whether they can handle big data processing. Depending on each parameter and requirements there are significant differences in group pattern discovery which leads to several challenges in finding the best group pattern algorithm.

Due to the complexity of spatio-temporal data itself as well as its continuously growing and changing volume there are several challenges that are worth noticing and which have also been re-occurring in the related literature. These challenges can be summarized as per below:

- **Stream analysis:** Growth of streaming technologies which generates high volumes of data require a new approach to extract valuable information from such data

- **Incremental discovery:** Many current applications (especially monitoring or surveillance) require the output results simultaneously while receiving and processing the streamed data which means the algorithms should be able to discover group patterns in an incremental manner.
- **Efficiency:** Streaming means huge amount of data generated in a very short time which leads to extreme computational cost of processing. The algorithms should then be able to output the groups efficiently.
- **Big data processing:** Huge amounts of spatio-temporal trajectories data require a distributed or parallel approach to handling the volumes and to reduce the computational overload
- **Parameter-free algorithms:** Many algorithms include a large number of parameters and each of them has a significant impact on the performance and the precision of the results. There is no standardized way of choosing the right set of parameters rather than by conducting experiments which is not an ideal situation
- **Group continuity:** Traveling together may not always mean the members of the group stay together during the lifespan of the group. This is particularly noticeable in human trajectories and in larger groups where members of the group can form sub-groups (Naserian et al., 2018)
- **Effectiveness:** The number of members of the group can vary and the effective algorithm should be able to understand the wider picture and report long-term groups rather than short term ones.
- **High Precision:** The challenge lies with discovering the correct groups rather than more groups which can make a difference in certain applications based on personalized services and targeting the wrong group may be worse than not addressing it at all.
- **Data mining theory and formalization:** Esling as well as many other researchers noticed the lack of formalization of data mining theory such as through solid mathematical foundation (Dodge et al., 2008; Esling and Agon, 2012; Zheng, 2015).

Many researchers highlighted the importance of readjusting the currently existing algorithms to extract trajectories coming from streamed data and in the incremental manner rather than from static/historical datasets. This was one of the reasons why the traveling companion algorithm was presented together with a data structure *traveling buddy* which allowed storing the object relationships instead of their spatial coordinates (Tang et al., 2012). Analyzing

and mining streamed data comes at a very high computational cost and processing it in an incremental manner means that there is a huge amount of data flow at a very short time and many algorithms do not scale up to big data (Esling and Agon, 2012). Therefore it is expected from the group discovery algorithms to be efficient (Naserian et al., 2018; Tang et al., 2012). One of the ways of boosting the efficiency is to handle spatio-temporal trajectory data via distributed, or parallel computing. There are several examples of modifying the existing algorithms to handle big data processing. Orakzai et al. in their work (Orakzai et al., 2016), amend the convoy pattern for distributed processing while Vatsavai et al. analyze various algorithms from the distributed computing point of view (Vatsavai et al., 2012).

In his work on time-series data mining, Esling highlights another important research trend and challenge at the same time which is parameter-free data mining (Esling and Agon, 2012). The majority (if not all of the reviewed algorithms) are hugely dependent on parameters/thresholds. These parameters affect both the performance of the algorithms and the precision of the results. Most of the discussed algorithms would include the size threshold and duration threshold (flock, convoy, swarm, traveling companion). Loose traveling companion includes additionally the frequency threshold and time-gap threshold. Given the example of the loose traveling companion setting up the size threshold higher results in a decrease of the time cost as it is easier to extract bigger groups. On the other hand, choosing the right value for the duration threshold depends on the type of the application and the other circumstances which can impact the number and precision of the identified groups. Taking as an example the closed locations with many changing groups of people such as airports or shopping centers setting up the duration threshold to a higher value such as days or weeks may result in completely different outcome to if it is set up to hours, or minutes. Some researchers attempted to create parameter-free models (Keogh et al., 2004) which are worth mentioning and ideally a perfect model should be parameter-free or have as few as possible so the results are not biased, nor in any way affected.

Specifically, in regards to group pattern discovery the challenge of defining a group has been taken by many researchers. Traveling together may not necessarily mean the members of the group will always stay together during their travel time and especially in human trajectories members of the group can split into sub-groups which require some flexibility in regards to how the continuity of the group is defined. One of the solutions as mentioned in the context of the parameters is to either modify the duration threshold or introduce a time-gap threshold allowing certain period of time when the members can stay apart. However, this is always subject to the right set up of the parameters which proves to be hard. Many state-of-art algorithms and clustering methods seem to ignore the fact that the group may split (Li et al., 2010; Zheng et al., 2013).

High precision and effectiveness of group discovery algorithms is strongly related with identifying the right groups rather than more groups and understanding the bigger picture. Similarly to the discovery of the sub-groups the correct groups identification is dependent on the right parameters (Naserian et al., 2018).

2.1 Statement of Problem

Taking into consideration all above challenges and the lack of a comprehensive classification of group pattern mining algorithms from spatio-temporal trajectories, there are several questions that can be answered in this dissertation to fill the gap. The three of key research questions in this context are following:

1. What are the commonalities of the existing algorithms for group discovery from spatio-temporal trajectories?
2. Can they be generalised to form a conceptual framework for group pattern mining from spatio-temporal trajectories?
3. What are the features of the optimal/"ideal" model that resolve the gaps in this research area?

These can be further split into sub-questions to support the research in this area:

1. What are all currently available algorithms for group discovery from spatio-temporal trajectories?
2. What are their definitions?
3. What are their features?
4. What are their similarities and differences?
5. What are currently known frameworks for group/data mining from spatio-temporal trajectories?
6. What are the challenges in this subject area?
7. How these challenges have been addressed so far?

2.2 Goals and Objectives

The goal of this dissertation is to contribute to the future research and development of group pattern discovery algorithms from spatio-temporal data by providing a conceptual framework for group pattern mining, followed by a comprehensive classification of existing models. Additionally, since this dissertation is considered as the step towards developing a novel algorithm in the future, the result of this extensive literature review brings the concept of an "ideal" model for group discovery from spatio-temporal trajectories. The framework includes both regular and big data as well as data stream processing models which to the best of author's knowledge is the first attempt of presenting both approaches in this context. In addition to this the current literature does not present any detailed publication specifically on the classification of group pattern discovery algorithms from spatio-temporal trajectories. The closest to this concept are the works of Zheng (Zheng, 2015), Alturi (Alturi et al., 2017) and Dodge (Dodge et al., 2008). Nevertheless, Zheng presents the whole trajectory data mining process where group pattern discovery or "moving together patterns" (as mentioned in that paper) is only briefly explained among three other trajectory pattern mining approaches as the final element in the discussed paradigm. Similarly, Alturi focused on the general spatio-temporal data mining process from the perspective of problems and methods while Dodge, whose work is the closest to this approach, provided a framework and classification of movement patterns from the perspective of the movements behaviours. Furthermore, the latter analyses all movement patterns including spatial, temporal as well as spatio-temporal solutions and was published in 2008 and therefore does not include the last 10 years of the research done in the spatio-temporal area which forms the basis of the proposed dissertation.

In order to achieve the goal of this dissertation project, it has been split into smaller objectives which are following:

- extracting definitions of the group pattern discovery algorithms from available literature
- extracting models of the algorithms from the available pseudo-codes and attached documentations
- grouping the definitions and models based on commonalities
- drafting a conceptual framework based on identified processes
- classifying the algorithms based on the framework
- defining the "ideal" model features
- indications for future research

Chapter 3

Methods

3.1 Methodology

In order to achieve the dissertation goals, an extensive literature review has been conducted. Since a significant majority of the published work in this and related research areas do not include the source code, nor the original datasets, this work is primarily based on secondary sources from the existing publications in the discussed subject. The review has been done based on literature search with the following key words and phrases: group pattern mining from spatio-temporal trajectories and its variations such as spatial-temporal, spatial temporal, spatiotemporal, SpatioTemporal, spatio-temporal data, spatio-temporal data mining, including movement trajectory, group pattern discovery, moving object(s) but also related areas such as spatial, temporal, space time, spacetime, geospatial, pattern mining, group pattern mining, data mining, parallel trajectory mining, distributed trajectory mining, distance measures, data indexing, sequence matching, similarity measures, stream analysis, temporal analysis, time series, trajectory, data stream, clustering. In order to create own framework and taxonomy structure a separate search on key words for frameworks and taxonomies in this context was made. This extensive search resulted in over 100 peer-reviewed papers from renowned journals and book chapters ranging from 1996 to 2018. However, only around 60% of them is related to spatio-temporal data in various aspects including pre-processing, data mining, pattern mining and out of these 20+ key algorithms were extracted that can be classified in the research area of group pattern mining from spatio-temporal trajectories. This is by far the most extensive research in this topic compared to other papers (Dodge et al., 2008; Feng and Zhu, 2016; Kisilevieh et al., 2010; Shekhar et al., 2015; Zheng, 2015)

This dissertation takes a form of a survey/taxonomy of the existing group discovery models from spatio-temporal trajectories. In order to be able to perform a comprehensive classification of these models, a conceptual framework has been proposed that encapsulates

all the reviewed features and processes leading to group discoveries from spatio-temporal trajectories. This is based on a top-down approach by formulating a general concept and overview of the group pattern discovery from spatio-temporal trajectories and then refined based on the result of a bottom-up approach taken to define and classify the algorithms by decomposing their elements into parameters, influencing factors, etc.

A revision of the existing frameworks and approaches to data mining in spatial or spatio-temporal trajectories leads to a conclusion that these frameworks are either simplified and not including the details of the whole process which means they cannot be generalized to include other models, or they are not specifically designed for group pattern mining. Three most commonly used frameworks for group pattern discovery which are similar if not the same in their essence are: filter-refinement, partition and grouping and clustering and intersection. Filter and refinement are used for the classical algorithms such as flock and convoy and with certain amendments in more recent patterns such as traveling companion (Tang et al., 2013) but do not take into consideration the input data which may require for instance big data processing. Partition and grouping is a similar approach used in trajectory clustering (Lee et al., 2007) while clustering and intersection are mentioned in several publications on various approaches (Gudmundsson and Horton, 2016; Jeung et al., 2010; Zheng, 2015). In *Spatiotemporal Data Mining: A Computational Perspective*, Shekhar et al. introduce a framework for the data mining from spatio-temporal trajectories (Shekhar et al., 2015), however this process does not include different input data such as static, streamed, or big data which change the way of pre-processing data in order to lead to group discovery. Another attempt to introduce a unified theoretical framework for data mining in general was made by (Khan et al., 2013) which included key elements such as data sets, cluster sets set of rules and visualization, however again did not differentiate between the data type and data size, nor allowed other methods than clustering (which are for instance a case of the original flock algorithm which introduced disc based group formation rather than density clustering known from later approaches).

3.2 Group and movement parameters

In order to identify and specify the group patterns that are the subject of this paper, certain parameters have been established which allow to distinguish them. Definitions and key concepts concerning moving objects and their trajectories are further discussed in 4.1. However, in order to differentiate these patterns from others, the initial top-down approach to classify them has been based on the group and movement parameters. The first and most obvious criterion is the size parameter of the group and in the context of groups, the focus

Table 3.1 Classification - Group Patterns

Model	Dimension		Primitive	Primary derivatives				Secondary Derivatives			
	Spatial	Temporal	Position	Time-gap	Distance	Duration	Lifetime	Density	Radius	Similarity	Frequency
flock	✓	✓	✓			✓	✓		✓		
convoy	✓	✓	✓		✓		✓	✓		X	
swarm	✓	✓	✓		✓			✓			
GP	✓	✓	✓		✓		✓	✓			
TC	✓	✓	✓		✓	✓		✓	buddy	X	
LC	✓	✓	✓	✓	✓	✓		✓	buddy	X	
MC	✓	✓	✓					✓		✓	
AF	✓	✓	✓						✓		
EVOCO	✓	✓	✓	✓				✓			✓
WCGM	✓	✓	✓	✓	✓	✓	✓	✓			✓
LTCP	✓	✓	✓		✓	✓	✓				✓
WCLTCP	✓	✓	✓	✓	✓	✓	✓				✓
leadership	✓	✓	✓						✓		
convergence	✓	✓	✓						✓		
encounter	✓	✓	✓						✓		
VF	✓	✓	✓	✓*					✓		
meet	✓	✓	✓						✓		
MEMO	✓	✓	✓						✓		
DC	✓	✓	✓		✓						
BFE	✓	✓	✓	✓	✓						
MF	✓	✓	✓						✓		
LGMP	✓	✓	✓	✓	✓	✓					
IG	✓	✓	✓		✓		✓	✓	buddy		

GP - gathering pattern, TC - traveling companion

LC - loose companion, MC - moving cluster, AF - adaptive flock

EVOCO- evolving cluster, WCGM - weakly consistent group movement

LTCP - loose traveling companion pattern

WCLTCP - weakly continuous loose traveling companion pattern

VF - variable flock, MEMO - meeting of moving objects

DC - distributed convoy, BFE - basic flock evaluation

MF - moving flock, LGMP - Loose Group Movement Pattern

IG - incremental gathering

is on algorithms concerning groups sizes > 1 . The movement parameters are derived from kinesiology which defines the physical movement of objects which matches the group pattern discovery from spatio-temporal trajectories as it is understood in the context of this work as moving point objects trajectories. This classification has roots based on two concepts that were extended for the purposes of this work. One called REMO (RElative Motion) which compares the motion attributes of moving point objects over time and space (Laube and Imfeld, 2002). The second is following Dodge's classification who defined the set of moving parameters by grouping them into primitive, primary and secondary derivatives (Dodge et al., 2008) based on the spatial, temporal and spatio-temporal dimensions. This set of movement parameters have been further extended and modified to allow the classification of the patterns that are presented in this work. Primitive parameters in spatial dimension are simply the (x,y) position of the moving point object. In the context of the temporal dimension these are the instances and intervals of time. A primary derivative of spatial dimension is the function of position such as distance, and then a secondary derivatives which is the function of distance are density, radius and similarity parameters. In regards to temporal dimension, the primary derivative is duration, lifetime of a group and window while the secondary derivative in this context is a frequency which is a function of duration. In spatio-temporal dimension (x,y,t) the primary derivatives are speed and velocity both being functions of (x,y,t) while the secondary derivatives include for instance acceleration which is the function of speed.

The Table 3.1 shows the outcome of the classification based on spatial and temporal dimensions as well as the thresholds/parameters resulting from them. These algorithms will be further revised based on the KDD derived framework as shown in 3.1.

3.3 Framework

The framework presented in Figure 3.1 is derived from a traditional KDD process (Tan et al., 2014) for knowledge discovery in databases and is modified to handle distributed computing models and data stream. It provides a core structure for further analysis and classification of the discussed algorithms.

While results validation and visualization are beyond the scope of this work due to resource limitations, the key focus for describing and defining group discovery algorithms from spatio-temporal trajectories will be on the pattern mining stage. Input data, its type and size as well as pre-processing models play an important role in algorithms' design which will be further discussed in Section 5. In this model there are three different pre-processing stages which are dependent on the input data type and data size. The algorithms can be classified based on whether they scale up and are compatible with streaming data or not and whether

they require a distributed or parallel programming or not. Next step is the pre-processing stage which includes all forms of clustering or filtering as per aforementioned frameworks and then again depending on the chosen pre-processing method the algorithms can be classified accordingly. The group discovery algorithm would then include all sort of parameters such as duration, density, group size, frequency, time-gap, etc. which eventually lead to analysis and validation of the results based on the selected parameters. Visualization is an additional UI layer which is beyond the scope of this dissertation.

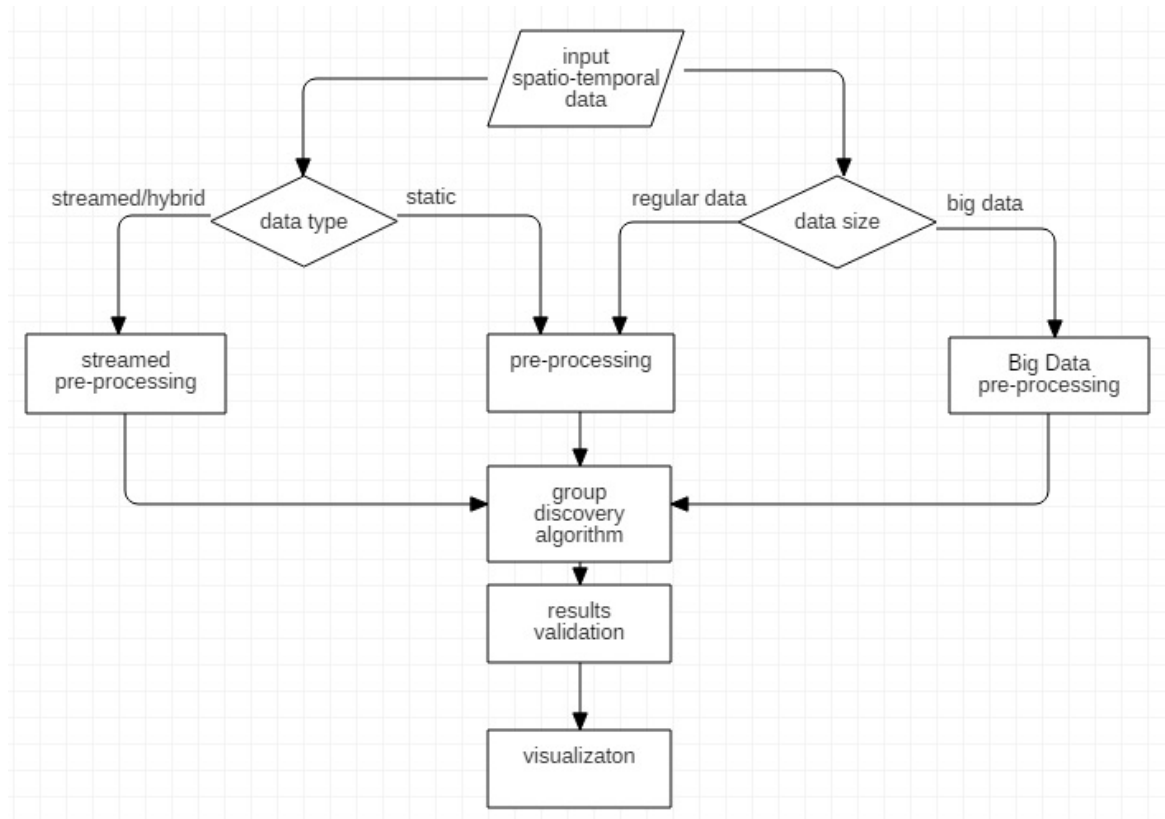


Fig. 3.1 Framework for group pattern discovery from spatio-temporal trajectories

Chapter 4

Classification

4.1 Definitions and Key Concepts

This section includes a set of definitions that allow certain level of standardization and generalization between various group discovery algorithms from spatio-temporal trajectories. The elements of these definitions have been reoccurring in the discussed algorithms sometimes with different nomenclature but performing similar functions, or representing similar concepts. Since a vast majority of the algorithms use clustering to identify moving objects, the definitions include cluster and cluster-sets accordingly. In addition to this and as mentioned on several occasions each of the algorithms use a different set of thresholds. The most common ones will be defined in this section and developed further in section 4.4. The key notations discussed in this section have been presented in Table 4.1.

Definition 1. *Moving object*

In the context of group pattern mining from spatio-temporal trajectories the focus is on the discovery of a group of moving objects that travel together over certain period of time (Jeung et al., 2008; Tang et al., 2012; Zheng, 2015). The most common differentiators of these groups are group size (number of objects in the group), shape/density of the group and the lifetime or duration of the pattern (Zheng, 2015). However, this list of differentiators can be extended which will be further discussed in thresholds' definitions. The moving objects in this subject area are primarily understood as Moving Point Objects (Laube et al., 2004) rather than Moving Regions or Line abstractions presented in the wider context of spatial, temporal or spatio-temporal data mining (Güting and Schneider, 2005; Huere and Santos, 2011). Moving Region includes the physical object as well as the surrounding area and which can consist of missing elements. This can be for instance an abstraction of oil spills, hurricanes or epidemic diseases (Güting and Schneider, 2005; Zheng, 2015). A line is an abstraction for moving in space which can represent rivers, or roads (Güting and Schneider, 2005). A point

Table 4.1 Table of Notations

Notation	Definition
O	moving object
O_{set}	set of objects
V/V_{next}	candidates/current candidates
Tr	trajectory
Ts	trajectory stream
C	cluster
t	time-slot index/snapshot
δ_s	size threshold
δ_d	duration threshold
δ_g	gap threshold
δ_{dn}	density threshold
δ_{ds}	distance threshold
δ_f	frequency threshold
δ_r	radius threshold
δ_{sm}	similarity threshold
P	group (pattern)

represents an object which has its location in time and space (from a spatio-temporal data point of view). Moving Point Object is a frequent and most basic representation of physical objects (Huere and Santos, 2011; Laube et al., 2004). In the respective literature the moving points are representing animals (Gudmundsson, 2004; Wang and Ram, 2015), objects (Jeung et al., 2008, 2010), or humans (Naserian et al., 2018; Tang et al., 2012; Zheng et al., 2013). The Figure 4.1 shows a 3-D graphical representation of points, lines and regions to show the difference between them.

While in the general visualization of the moving objects they are represented as single points as, in particular case of group pattern mining, the group will consists of a number of moving points **O** where the number of moving objects is $N > 1$ (Dodge et al., 2008).

Definition 2. *Set of objects*

A set of objects $O_{set} = \{O_1, O_2, \dots, O_n\}$ is a set of moving objects which can be a collection of object data from a database (Benkert et al., 2008; Gudmundsson and van Kreveld, 2006), or a result of the pre-processing/preparation stage where an initial object database is created that then becomes an input for further analysis (Khoshahval et al., 2017; Naserian et al., 2018). While the majority of the discussed algorithms will be using a trajectory database rather than a moving object database which contains a travel history of the moving object rather than the current location of the moving object, certain key group discovery algorithms in this research area such as flock and its variations has been defined as

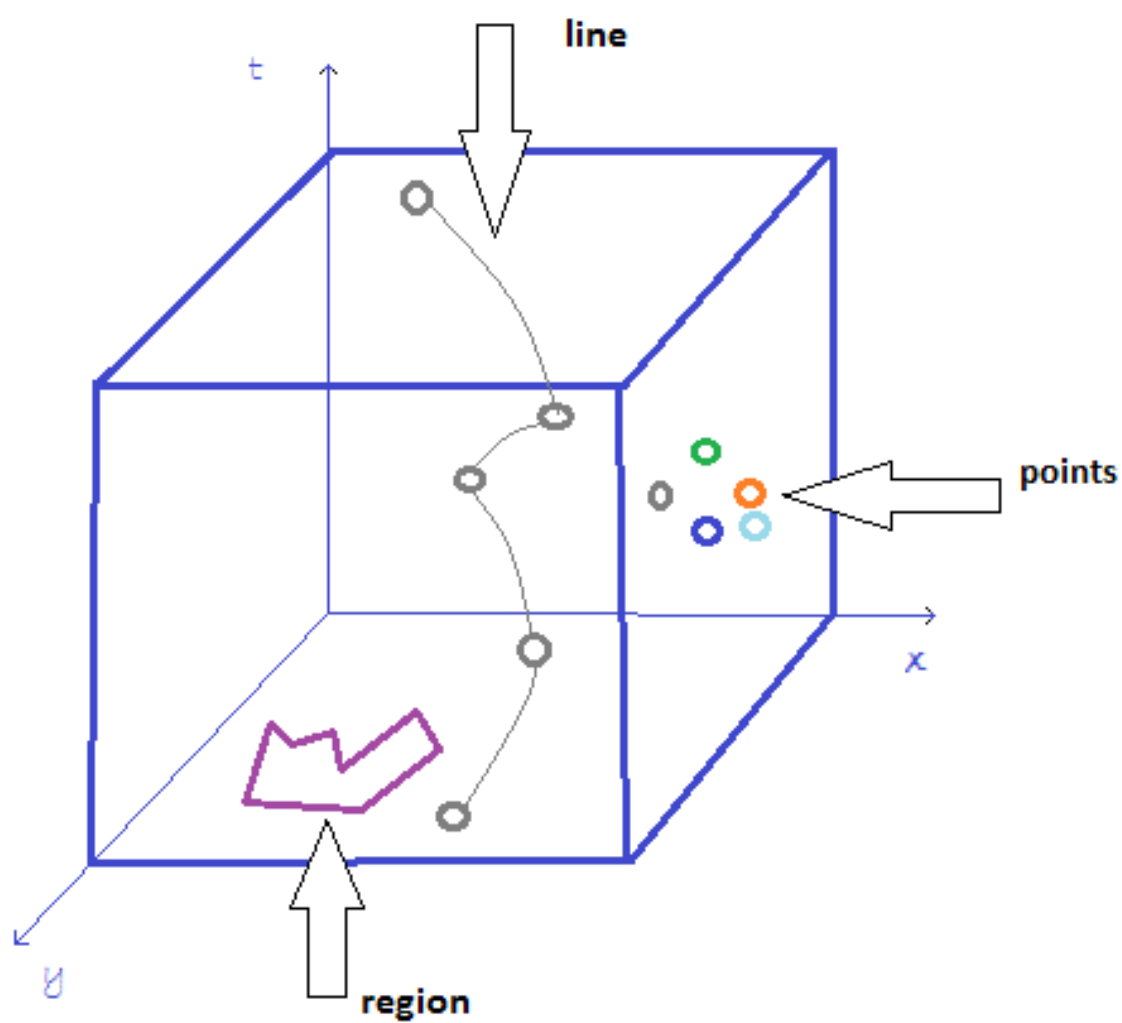


Fig. 4.1 3D visualization of Point Objects, Region and Line

taking input data from moving object databases (Benkert et al., 2008; Zheng et al., 2014). In this case the set of objects will be mentioned as referring to object set from moving object databases. A significant majority of algorithms are specifically focused on trajectory databases (Jeung et al., 2008, 2010; Naserian et al., 2018; Tang et al., 2012). In this context the set of objects will be referred as resulting from the pre-processing stage and understood as a subset of the input data that has the potential of generating group pattern results (Patel, 2015). However, in the context of clustering and intersection and also when dealing with streaming input data approach as opposed to static database input, Tang et al. (Tang et al., 2012, 2013) also refer to the set of objects O_{set} as a collection of moving objects in the given snapshot, rather than a collection of objects from a database, or a subset of input data from the pre-processing stage. The snapshots are different to the time threshold for group discovery and for instance snapshots can be set up to every 10 minutes while group discovery can be monitored over a time of 50 minutes. Therefore, basing on the available literature, there are three key understandings of the set of objects which can be referred to as 1) a set of moving objects directly from moving objects database, 2) a subset of input data resulting from the pre-processing stage, or 3) a collection of moving objects in the given snapshot which is most relevant for streaming input data models. Each of them will be specified accordingly to the discussed algorithms models and approaches.

Definition 3. *Candidates/Current Candidates*

V is the set of potential moving objects that can form a group pattern. The set of V is the initial group discovery which is usually created after the initial pre-processing stage provides a general database of clustered objects and their trajectories.. V_{next} is used to store temporary candidates at a current time which can be added to the V if they match the criteria. V eventually becomes a group pattern if after final refinements matches all required thresholds.

Definition 4. *Trajectory*

Unlike moving object databases which contain current location of the moving object, trajectory database contains a traveling history of a moving object (Zheng, 2015). In this context Tr is understood as a static trajectory database as opposed to stream trajectories which are denoted as Ts and explained in the following definition. A database trajectory is a sequence of geo-locations with corresponding timestamps in spatio-temporal environment such as $Tr = \{ \langle p_1, t_1 \rangle, \langle p_2, t_2 \rangle, \dots, \langle p_n, t_n \rangle \}$ where each element $\langle p_i, t_i \rangle$ refers to a moving object at a location p_i and a timestamp t_i . Each of these elements are sorted by timestamps, following $t_j < t_k$ if $1 \leq j < k \leq n$ (Feng and Zhu, 2016; Zheng et al., 2014). A geo-location is usually represented as tuples of $\langle \text{longitude}, \text{latitude} \rangle$ when coming from GPS device data, or $(t, (x, y, z))$ with x - y - and z - coordinates of the moving object O at the time t .

For instance, RFID uses tags or readers (Feng and Zhu, 2016). A trajectory of the moving objects is most commonly represented by a polygonal line (Benkert et al., 2008; Zheng et al., 2014).

Definition 5. Trajectory stream

Trajectory stream Ts is understood as a sequence of snapshots $\{Ts_1, Ts_2, \dots, Ts_n\}$ where each snapshot of an object O contains spatial and temporal coordinates of this object at the given snapshot. Each Ts_n can be denoted as a positional record of a moving object O resulting in a pair of (O^j, Ts_i^j) where Ts_i^j is the location of the object O^j at time instant t_i in a 2D space (Mao et al., 2018). The key difference between this approach and a database trajectory data is that the trajectory stream data is added in an incremental manner rather than from pre-existing historical trajectory data. In case of trajectory data stream the positions of moving objects can be received in real time at each snapshot in a form of data stream which is dynamic and different to static trajectory data where this positions are fixed (Yu et al., 2013).

Definition 6. Cluster

In the majority of cases, clustering is used in the pre-processing step (Bialkowski et al., 2016; Jeung et al., 2008; Li et al., 2010; Naserian et al., 2018; Orakzai et al., 2016; Zheng et al., 2013) resulting in a cluster database of cluster-sets $C = \{C_1, C_2, \dots, C_n\}$, where each of the clusters is extracted at the given time-slot which is also called a snapshot clustering (Tang et al., 2013; Zheng et al., 2014). Since many of the algorithms models follow a density based clustering, a snapshot cluster in this context is a group of objects which are density-connected to each other at a given snapshot (Feng et al., 2015; Kisilevieh et al., 2010; Tang et al., 2013; Zheng et al., 2013, 2014).

Definition 7. Time-slot/snapshot

Time-slot is understood as the time interval or a snapshot $t \in \{1, \dots, T\}$, where T can vary from a short time of a minute to longer period of times depending on the application. The snapshot is particularly relevant for streaming data trajectories.

Definition 8. Thresholds

Thresholds are the most significant differentiators between various algorithm approaches in group pattern mining from spatio-temporal trajectories. They also generate significantly different outcomes within the same algorithms depending on threshold set up. Zheng in the survey on trajectory data mining (Zheng, 2015) mentions three key differentiating thresholds which are the group size, density or shape of the group and the duration of the pattern. Further review of the algorithms from this research area shows that many other thresholds play an important role in group discovery. The consequences of different thresholds set up will be

further discussed in section 4.4 while here the definitions will be provided. The significant thresholds appearing in the subject literature are following:

- **Size threshold** δ_s is the size of the targeted group to be discovered from pattern mining
- **Duration threshold** δ_d is the minimum lifetime of a group.
- **Gap threshold** δ_g refers to the time-gap between cluster-sets which breaks the strict k-consecutive continuity of the group and allows the group members to leave the group and re-join within the specified gap-threshold.
- **Density threshold** δ_{dn} is derived from the density-based clustering method that is commonly used in the pre-processing stage (DBSCAN) where for each point of a cluster, the neighborhood of a given radius has to contain at least a minimum of points so the density of this neighborhood needs to exceed the density threshold (Ester et al., 1996)
- **Distance threshold** δ_{ds} is related to the same density-based clustering method and defines the shape of the neighborhood as a distance function of two points p and q $\text{dist}(p,q)$ (Ester et al., 1996)
- **Frequency threshold** δ_f defines the minimum time-slots when the group stays together
- **Radius threshold** δ_r is specific to flock group of patterns where at least m number of moving objects O are found within a circular region of constant radius $\delta_r > 0$ and heading in the same direction at some point of time
- **Similarity threshold** δ_{sm} defines the minimum of overlapping moving point objects in any two consecutive clusters and the value of $\delta_{sm} \in [0,1]$. Given the example of $\delta_{sm} = 0.7$, it is required that 70% of the clusters are required to overlap.

Definition 9. *A group*

The group is understood as the result of group pattern discovery so it is a set of moving objects that fulfills the search criteria (i.e. matches the required thresholds). In order to classify and define movement patterns and the algorithms in the context of groups certain relationships, or behaviours resulting from thresholds criteria will be taken into consideration. The relationships allow, for instance, to distinguish groups from cohorts where cohorts are understood as a group of moving objects (usually people) who share a defining characteristic over a period of time (Dodge et al., 2008) which might be also related to a participation in

certain events such as parades, or celebrations. Certain group pattern algorithms from spatio-temporal trajectories such as gathering (Zheng et al., 2013, 2014), or meeting (Gudmundsson and van Kreveld, 2006) are particularly focused on these differentiating events and bring additional and interesting options to the algorithm models such as a possibility for a member of a group to leave and rejoin the group. The understanding of the relationships between the moving objects is a part of the interpretation of the patterns rather than defining them (Dodge et al., 2008). However, many of the relationships are included in the thresholds present in the algorithms such as the time the member can stay with the group, the time gap the member is allowed to leave the group and then rejoin, etc. These relationships can also refer to a stationary versus moving group formation. While meet, or gathering patterns are an example of a stationary cluster where the objects stay together for a certain period of time, the flock, or convoy are examples of moving clusters where a set of objects stay together for a certain period of time while following the same trajectory. Further modifications such as loose traveling companion algorithm allow the objects to leave and rejoin the group while the group is traveling/moving.

Therefore, a group P is understood as a set of moving objects and including both stationary and moving clusters which fulfill the threshold criteria and where the number of members in the group $N > 1$.

4.2 Spatio-temporal Input Data

This section will provide information on spatio-temporal input data and its characteristics. This will be followed by further analysis of spatio-temporal data attributes, data size and data type and how these are reflected in the existing approaches to group pattern mining from spatio-temporal trajectories.

4.2.1 Data Attributes

Data attributes in the context of this work can be grouped into three different types: non-spatiotemporal attributes, spatial attributes, and temporal attributes (Shekhar et al., 2015). A sample of a spatio-temporal dataset is shown in Figure 4.2. The first type of the attributes is used to capture certain characteristics of objects which can be some forms of ID, reference, or name among others (Tan et al., 2014). The non-spatiotemporal attributes have a particular significance in predictive models such as next location prediction (Naserian et al., 2018). Any contextual information about the object and in particular in case of human trajectories such as gender, age, their travel plans and times as well as social information on preferences, groups

```
'data.frame':  313 obs. of  3 variables:
 $ Time: int  5863 5870 6017 6018 6034 6060 6176 6364 6366 6372 ...
 $ X    : num  63.9 64.3 64.1 64 64.4 ...
 $ Y    : num  19.4 20.1 19.7 19.8 20.3 ...
```

Fig. 4.2 Sample of spatio-temporal data from R *spacetime* package

they associate with, etc. is an extra layer on top of the spatio-temporal data mining. Spatial and temporal attributes are the ones differentiating moving object trajectory from classical data mining sets. Spatial attributes as shown in the given example 4.2 define the geolocation of the moving objects(s) such as longitude and latitude while temporal attributes include the timestamp of the moving object, snapshot, or the duration (Patel, 2015; Shekhar et al., 2015). There is no standardized approach to handling spatio-temporal data mining mostly due to complexity and multi-dimensionality of such data. One of the approaches to mine the spatio-temporal trajectories of moving objects is to convert them into traditional input columns (like in the given example) and use classical data mining methods to extract patterns or build predictive models. Another approach is to develop customized spatio-temporal tools including new statistics methods which have been further discussed by Shekhar in his survey on computational perspective of spatio-temporal data mining (Shekhar et al., 2015).

4.2.2 Data Type Source

While the literature categorizes spatio-temporal data into the following types: temporal snapshots, temporal models and events (Shekhar et al., 2015; Yuan, 1996), the focus of this dissertation will be on another approach to data type by looking at it from the data source perspective and dividing the algorithms based on their static and/or streaming data handling capacity.

Majority of the current algorithms for spatio-temporal group discovery is designed to work with static databases. These static databases handle spatial and temporal data and are optimized for storing and querying them. The data represents objects that are defined in a geometric space which is usually the 2-dimensional Euclidean space and the representation is in the form of simple geometric objects such as points, lines, or regions (polygons). Some databases are capable of handling more complex 3-dimensional objects. The stored data in the context of this dissertation is considered as a set of moving objects $\mathbf{O}_{\text{set}} = \{\mathbf{O}_1, \mathbf{O}_2, \dots, \mathbf{O}_n\}$ (Benkert et al., 2008; Gudmundsson and van Kreveld, 2006; Zheng et al., 2013) (which is further presented in Section 4.1). Another static input data can be a trajectory database which contains a traveling history of a moving object (Zheng, 2015). This traveling history is stored as a sequence of geo-locations with corresponding timestamps in spatio-temporal

environment such as $Tr = \{ \langle \mathbf{p}_1, \mathbf{t}_1 \rangle, \langle \mathbf{p}_2, \mathbf{t}_2 \rangle, \dots, \langle \mathbf{p}_n, \mathbf{t}_n \rangle \}$ where each element $\langle \mathbf{p}_i, \mathbf{t}_i \rangle$ refers to a moving object at a location $\mathbf{p}_i$ and a timestamp $\mathbf{t}_i$ when the location was captured. Each of these elements are sorted by timestamps, following $\mathbf{t}_j < \mathbf{t}_k$ if $1 \leq j < k \leq n$ (Feng and Zhu, 2016; Zheng et al., 2014). In order to increase the efficiency of correct group discovery many algorithms perform multiple scans of static data and are not always designed to handle the database updates and output results incrementally (Tang et al., 2013).

Different from traditional static data, data streams are dynamic, continuous and subject to many challenges derived from it (Jiang and Gruenwald, 2006). A data stream is an ordered sequence of instances which arrive in a timely order. Due to limitations resulting from computing and storage capabilities a data stream is usually read only once or few limited number of times. Mao et al. in his survey on challenges in trajectory stream clustering mentions several issues that streaming algorithms should resolve. These include incomplete data, unlimited volume as opposed to limited memory space, evolving trajectories, obsolete data as well as variable sampling rate (Mao et al., 2016).

Due to failures of physical devices that are also imprecise and prone to errors, the incoming data stream may be incomplete and therefore various methods of reducing the data uncertainty are needed. Some of these may include different filtering methods to deal with the noisy data, or map-matching. Unrestricted data volume is another challenge which requires methods to facilitate fast processing without the need of storing all details in memory. As it will be shown in the algorithm design in Section 5, many models that allow streaming data processing introduce new data structures which do not store all trajectory/object information but for instance certain references such as object relationships like in the *traveling buddy* (Tang et al., 2013), or summary of the incoming trajectory like in the *EF* example (Mao et al., 2016). When looking at moving objects, regardless of the chosen group pattern, their trajectories change over time and the groups can evolve so the streaming data models look at how to address these changes, how to refer the new possible candidates and predict whether they can become a group member. A data stream usually comes with a high speed and its data distribution changes with time. On top of that the sampling data may not always be correct due to dynamics of the streamed data environment and therefore there is a need to improve data precision by performing various methods among which a commonly used is a linear interpolation.

4.2.3 Data Size

While a significant majority of the discussed state-of-the-art group pattern algorithms from spatio-temporal trajectories is designed for being implemented on single machines, the spatio-temporal data is incessantly growing to the levels where a distributed/parallel approach is

needed to handle it. In addition to this, the aforementioned data streams require off-loading the memory in order to allow fast processing of data input at a high speed. One of the answers to the question on when the data is too big for a single machine requires looking at whether the entire raw dataset can be loaded into memory, or not. Another reason to consider distributed computing models in this area of research is that due to complexity of spatio-temporal data as well as a high cost of storing it, since 2008 the standard relational databases stopped providing spatio-temporal extensions as mentioned by Revesz (Revesz, 2010). There are spatial extensions available in various relational databases, however the spatio-temporal is not only huge in size but also multi-dimensional, with complex structures, relationships and behaviours. Looking at the models of algorithms as well as the experimental datasets that were used to perform analysis most of them were performed on single machines. The parameters of the machines vary from single to dual CPU with 2.13GHz -3.3GHz (Jeung et al., 2010; Naserian et al., 2018; Tang et al., 2013) and the main memory from 2GB to 16GB (Jeung et al., 2010; Li et al., 2010; Tang et al., 2012). The real datasets used for experiments are relatively small which is due to the access reasons. However, in order to evaluate the precision of the results bigger datasets were generated synthetically. These datasets usually track objects (such as taxis), animals, or less frequently human trajectories. The real datasets usually contain anywhere from 20 - 100 human trajectory data (Naserian et al., 2018), 50 - 50 object and animal data such as taxi, bus, truck and cattle (Jeung et al., 2010; Naserian et al., 2018; Tang et al., 2012; Vieira et al., 2009) and related to them spatio-temporal data records. The synthetic datasets can include 100,000 to more than 10 million data records (Tang et al., 2013; Vieira et al., 2009).

Even though the synthetic datasets might be large, a majority of these algorithms do not have a structure to process data in a distributed/parallel way and they include a lot of in-memory processing which is not scalable for big data spatio-temporal datasets.

Therefore, taking into consideration data size, the algorithm designs will be discussed in the context of their capability to handle distributed/parallel computing models, or single machine processing only which will be further presented in Section 5.

4.3 Pre-processing

This section will focus on the pre-processing stage for group discovery algorithm models and the current methods being used across various approaches. The pre-processing stage will be discussed in the light of various clustering methods used to extract meaningful data for further processing. The pre-processing stage varies depending on the previously mentioned data types and size. Static data pre-processing applies the clustering methods on all historical

trajectories stored in a static database whereas streaming pre-processing due to the dynamic nature of streaming data and the requirement for lowering the computational cost uses different forms of micro-clustering small parts of data. Similarly, distributed pre-processing requires a different approach to clustering the initial dataset as the size of data is too big to be processed in one go, especially that many clustering methods are complex and come with a high computational cost. Literature mentions between three and four key types of clustering methods used in spatio-temporal data mining (Folino and Spezzano, 2002; Mao et al., 2016; Shekhar et al., 2011, 2015; Vieira et al., 2009).

These include hierarchical methods, density-based clustering, partitioning and grid-based clustering. From the point of view of this work where distributed and streaming models are taken into consideration, this list can be extended as follows:

- **Connectivity/hierarchical models** : including hierarchical clustering (HCLUST), BIRCH, CURE which builds models based on distance connectivity.
- **Partitioning and centroid models**: including the k-means algorithm or k-medoids.
- **Density- based models**: with the most commonly used DBSCAN and OPTICS which define clusters as connected dense regions in the data space.
- **Grid-based models**: for example STING
- **Distributed models**: for example distributed hierarchical clustering, partitioning
- **Continuous/Incremental models**: such as micro-clustering (Li et al., 2004)
- **Parameter-free models**: such as PFClust (Mavridis et al., 2013)

Hierarchical clustering is a popular method as it takes no parameters, or is parameter-light and has been used in several patterns including swarm, or loose traveling companion. It can have a *bottom-up* and *bottom-down* structure which is agglomerative and divisive respectively. The difference is that in the first one each observation is assigned to an individual cluster and with next iterations pairs up with other cluster while the latter is based on spitting one big cluster with all observations and then performing a recursion. Density-based clustering is the most popular clustering method used, especially in convoy derived models and is widely discussed and referred in the related literature (Aung and Tan, 2012; Khoshahval et al., 2017; Shekhar et al., 2015). The most commonly used one is the DBSCAN (Ester et al., 1996) where given a set of points in space, the points are grouped together based on their density connection. Grid- based clustering such as STING as proposed by Wang (Wang et al., 2003) is popular with evolving or moving group patterns that are subject to continuous

changes and which implement continuous clustering with larger datasets were high speed and performance are essential. These algorithms such as moving objects (Jensen et al., 2007), adaptive flock (Folino and Spezzano, 2002), moving flock (Andres Oswaldo Calderon Romero, 2011), or moving clusters (Kalnis et al., 2005). Distributed clustering models are primarily modifications of the existing clustering methods that scale up to big data processing such as distributed convoy (Orakzai et al., 2016). Continuous, or incremental models use forms of micro-clustering (Mao et al., 2016). And lastly, a parameter-free approach is still an open research area with works by Keogh, or Mavridis (Keogh et al., 2004; Mavridis et al., 2013) presenting the parameter-free, or parameter-light methods of clustering data. As surveyed by Zheng, the research in this area can be divided into moving object cluster discovery, trajectory clustering as well as trajectory clustering of common paths for a group of moving objects (Zheng, 2015). The approach to moving object clusters focuses on individual moving objects and their finding similar clusters based on their behaviours and patterns, whereas the trajectory clustering is focused on geometric approach to cluster trajectories. One of the biggest challenges in this area is to define the similarity between two object trajectories. Methods that are used include Longest Common Subsequence (LCSS), Dynamic Time Warping (DTW), Edit Distance on Real Sequence (EFR) or Edit distance with Real Penalty (ERP). These methods have been successfully used by many group pattern algorithms such as loose traveling companion (Naserian et al., 2018), swarm (Li et al., 2010), convoy (Jeung et al., 2010), or moving clusters (Kalnis et al., 2005). These methods have been implemented with various frameworks such as *partition-and-group framework* (Lee et al., 2007), *micro-and-macro-clustering* (Agrawal and Srikant, 1995), *clustering-and-intersection* (Tang et al., 2013), *clustering-and-union* (Kalnis et al., 2005), or *filter-and-refinement* (Jeung et al., 2010). All of these frameworks are further used after the processing stage to identify potential candidates which will be further discussed in algorithms design Section 5.

Since the main focus of this area of research is the group discovery, many researchers leave their pre-processing clustering methods open (Naserian et al., 2018; Tang et al., 2013; Zheng et al., 2014). However, based on the conducted experiments certain methods were chosen over other. Convoy pattern (Jeung et al., 2010) is most commonly using DBSCAN density based clustering method while loose traveling companion (Naserian et al., 2018) and travel and gathering (Zheng et al., 2014) preferred hierarchical clustering.

4.4 Thresholds

The group discovery outcome varies significantly depending on the set of thresholds that are chosen to define the group. Size threshold δ_s , Duration threshold δ_d , Gap threshold δ_g ,

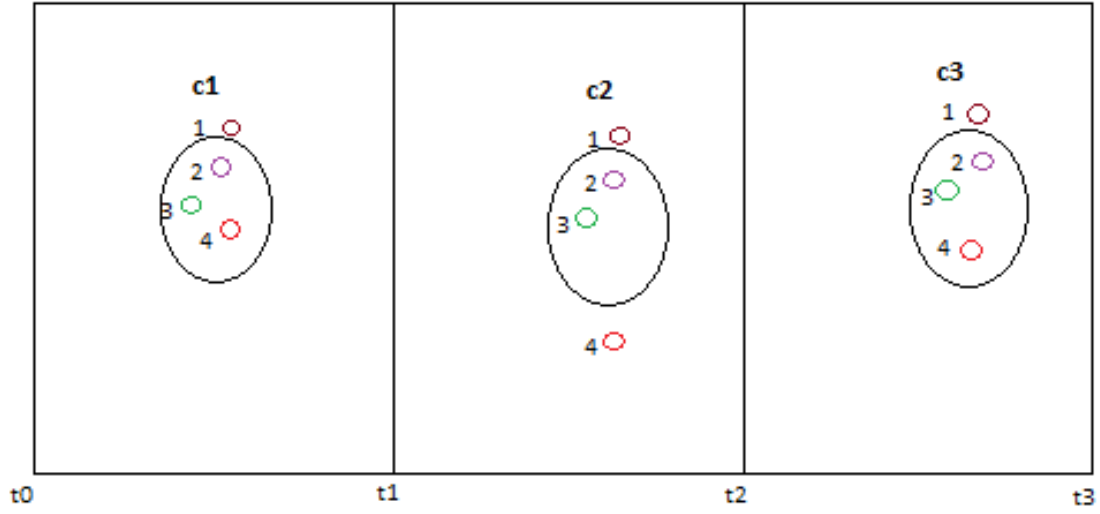


Fig. 4.3 Radius, Density and Gap thresholds

Density threshold δ_{dn} , Frequency threshold δ_f , Radius threshold δ_r , Distance threshold δ_{ds} and Similarity/Overlap/Integrity threshold δ_{sm} are the most commonly used in the discussed research area. Taking into consideration various environments such as indoor, or outdoor spaces as well as different moving objects (animals, objects, humans) these thresholds can be a hit or miss for group discovery.

Figure 4.3 shows differences in pattern discovery resulting from radius, density and gap thresholds set-up over a period of $T = 3$ timestamps with a group size $N = 2$. Basing on the the disc radius threshold and the group of 2 objects over 3 consecutive time-stamps, the flock pattern would discover objects $\{o_2, o_3\}$ would be considered as a group. Object o_1 should intuitively be also a part of the group as it is density connected to the other members. $\{o_1, o_2, o_3\}$ would be discovered as a part of convoy mining process as these are density connected members over a period of 3 consecutive timestamps. However, looking at the object o_4 it could potentially be a part of the same group but because it does not belong consecutively to the group and only during 2 out of 3 timestamps, then this object would not be discovered as a part of the convoy pattern mining. Different from convoy approach, swarm pattern mining allows members to be a part of the group during non-consecutive periods of time allowing them to disconnect and rejoin and therefore swarm would consider $\{o_1, o_2, o_3, o_4\}$ as a part of the moving objects group.

Nevertheless, allowing non-consecutive period of time may not always be the right solution as it assumes the members were together at the starting point and then disconnected and reconnected. Whereas sometimes different members may join the group later during

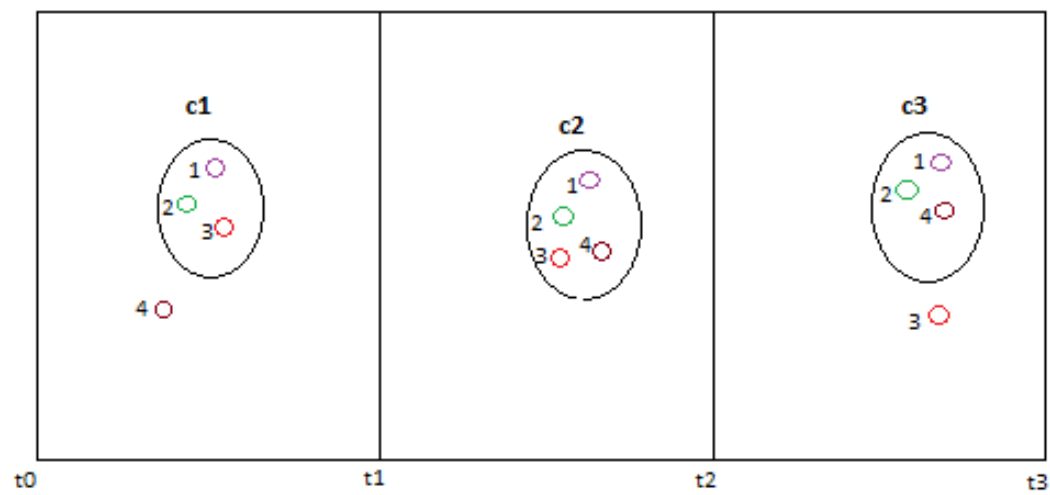


Fig. 4.4 Similarity/Overlap/Integrity threshold

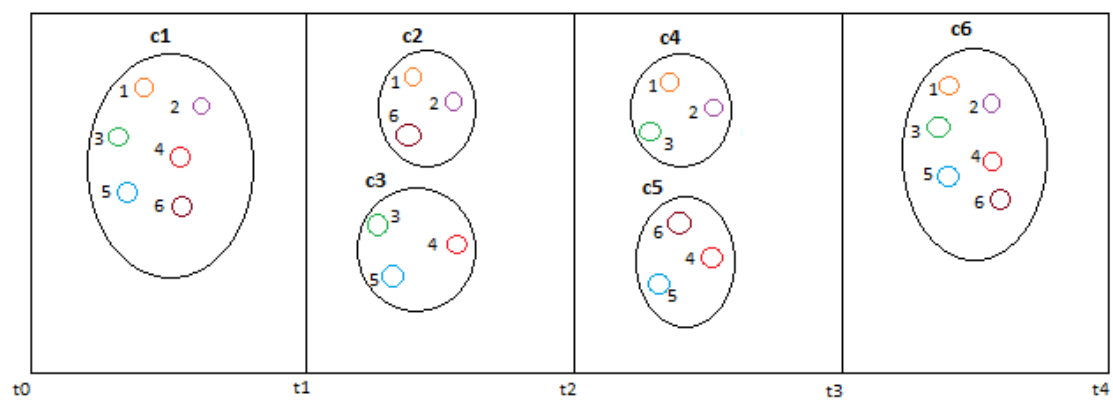


Fig. 4.5 Frequency, Similarity and Gap thresholds

the group lifetime. Figure 4.4 shows a different approach using similarity/overlap/integrity threshold as an example. Given the time threshold of $T = 3$ and the group number $N = 3$ and the similarity threshold $\delta_{sm} = 60\%$, the convoy pattern would not discover any group, simply because there is not any group of 3 members that are density connected together over 3 consecutive timestamps. The similarity threshold of 60% allows to discover that $\{\mathbf{o}_1, \mathbf{o}_2, \mathbf{o}_4\}$ are a cluster as the similarity threshold is fulfilled between $\mathbf{c}_2$ and $\mathbf{c}_3$ cluster-sets.

However, there are more criteria to extract groups and differentiate them from sub-groups. Looking at the Figure 4.5 and given the group size $N = 2$, time-gap = 1, $T = 4$ and frequency = 2 with an additional parameter set up to 2 which is taken from EVOCO and WCGM approaches and which defines continuity of the clusters basing on how many persistent member there are various groups that can be discovered. Following convoy pattern $\{\mathbf{o}_1, \mathbf{o}_2, \mathbf{o}_3, \mathbf{o}_4, \mathbf{o}_5, \mathbf{o}_6\}$ as well as $\{\mathbf{o}_1, \mathbf{o}_2\}$ and $\{\mathbf{o}_4, \mathbf{o}_5\}$ would be considered convoys. The WCGM pattern would discovery would discover the same groups with an exception that $\mathbf{o}_6$ would not be considered in the biggest group as it does not fulfill the restriction of the maximum time it can stay disconnected from the group. Contrary to these approaches, a WCLTCP pattern would only discover one group of $\{\mathbf{o}_1, \mathbf{o}_2, \mathbf{o}_3, \mathbf{o}_4, \mathbf{o}_5, \mathbf{o}_6\}$ treating the other groups as sub-groups of this group.

4.5 Taxonomy

The classification based on movement and group parameters as presented in Table 3.1 has been mapped with the elements of the conceptual framework presented in Figure 3.1 which results in the classification of group pattern mining algorithms from spatio-temporal trajectories presented in Table 4.2. The algorithms have been classified based on the input data type and data size, the pre-processing methods and movement parameters that are implemented in the algorithms models.

Table 4.2 Taxonomy - Group Patterns

Model	Input Data		Pre-processing	Primary derivatives					Secondary Derivatives		
	Data Type	Data Size	Clustering	Gap	Distance	Duration	Lifetime	Density	Radius	Similarity	Frequency
flock	static	SM	disc			✓	✓		✓		
convoy	static	SM	DBSCAN		✓			✓			
swarm	static	SM	DBSCAN		✓			✓			
GP	static	SM	DBSCAN		✓		✓	✓			
TC	stream	SM	DBSCAN		✓	✓		✓	buddy	X	
LC	stream	SM	DBSCAN	✓	✓	✓		✓	buddy	X	
MC	static	SM	DBSCAN					✓		✓	
AF	stream	parallel	SPARROW						✓		
EVOCO	stream	SM	DBSCAN	✓				✓			✓
WCGM	static	SM	DBSCAN	✓	✓	✓	✓	✓			✓
LTCP	static	SM	HCLUST		✓	✓					✓
WCLTCP	static	SM	HCLUST	✓	✓	✓	✓				✓
leadership	static	SM	disc						✓		
convergence	static	SM	disc						✓		
encounter	static	SM	disc						✓		
VF	static	SM	disc	✓*					✓		
meet	static	SM	disc						✓		
MEMO meet	static	SM	disc						✓		
DC	static	distributed	partition		✓						
BFE	stream	SM	disc	✓	✓				✓		
MF	stream	SM	disc						✓		
LGMP	static	SM	DBSCAN	✓	✓	✓		✓			
IG	stream	SM	DBSCAN		✓		✓	✓	buddy		

GP - gathering pattern, TC - traveling companion

LC - loose companion, MC - moving cluster, AF - adaptive flock

EVOCO- evolving cluster, WCGM - weakly consistent group movement

LTCP - loose traveling companion pattern

WCLTCP - weakly continuous loose traveling companion pattern

VF - variable flock, MEMO - meeting of moving objects

DC - distributed convoy, BFE - basic flock evaluation

MF - moving flock, LGMP - Loose Group Movement Pattern

IG - incremental gathering

Chapter 5

Algorithm Design

This section focuses on the group pattern mining algorithm design from spatio-temporal trajectories. Depending on the previously discussed data attributes, size and type as well as the set of thresholds/parameters, the algorithms are represented by different models. All the presented models are straightforward approaches aiming at discovering all groups matching given criteria and their generic versions will be presented in pseudo-codes. Firstly, a generic model from trajectory static database will be presented, followed by streamed input data model to address the incremental discovery and stream analysis as well as distributed/parallel model for big data processing. All these generic models will be further discussed basing on existing specific approaches resulting from the previously discussed classification criteria on data attributes, data type, data size and thresholds selection. The following section 6 will be concluded with an attempt to sketch a conceptual framework of a novel model for group discovery from spatio-temporal trajectories based on the outcome of algorithm design analysis of all mentioned models that could be potentially developed in the future. The models presented in this section are grouped in the following categories resulting from the classification based on the input data type and size:

- **static data examples:** this group is dominant as the majority of currently available algorithms in this research area use a static database and follow a more traditional KDD process such as flock, convoy, swarm, gathering, leadership, convergence, encounter, meet, MEMO, moving flock, group pattern (Apriori Group Pattern), moving convoy.
- **streamed data examples:** the development of this group became popular in the last few years and include loose and traveling companion models with a traveling buddy to store a reference rather than data for fast processing, LTCP and WCLTCP, notions of a continuous discovery of moving clusters (Jensen et al., 2007), BFA(basic flock evaluation), or EVOCO (can be used for both static and streaming)

- **distributed/parallel examples:** Distributed algorithms in this research area are currently in minority with few examples such as distributed convoy by Orakza (Orakzai et al., 2016), adaptive flock by Folino (Folino and Spezzano, 2002), or grid and spatial index methods such as TrajStore (Cudre-Mauroux et al., 2010) for distributed trajectory mining rather than a group discovery

5.1 Static data algorithms

The generic model from trajectory static database is presented in Algorithm 1. It takes the input of a static, historical trajectory data as well as a set of thresholds. It is the most commonly represented within the group patterning discovery algorithms from spatio-temporal trajectories. This algorithm model is relevant to majority of the classical algorithms such as convoy (Jeung et al., 2008, 2010), or swarm (Li et al., 2010), as well as gathering (Zheng et al., 2014), or group pattern (AGP) (Wang et al., 2006). Flock-based patterns such as "classical" flock (Benkert et al., 2008; Gudmundsson, 2004; Laube and Imfeld, 2002), moving flock (Andres Oswaldo Calderon Romero, 2011), leadership (Gudmundsson, 2004; Laube and Imfeld, 2002; Laube et al., 2004), convergence (Gudmundsson, 2004; Laube and Imfeld, 2002; Laube et al., 2004), encounter (Gudmundsson, 2004; Laube et al., 2004), meeting of moving objects(MEMO) (Aung and Tan, 2012) and meet (Gudmundsson, 2004; Laube and Imfeld, 2002; Laube et al., 2004) can also be grouped into this static model approach with one exception concerning the group formation. Unlike all previously mentioned models in this group that apply different methods of clustering to extract the group members, the flock-based patterns use a predefined, circular region with a constant radius to identify the members of the group, rather than via other clustering methods.

These static group discovery models can be split into three core stages as shown in the generic Algorithm 1 model:

1. pre-processing(clustering/region) (lines 1-2) // finds all possible combinations from trajectory (snapshot clustering)
2. extension of the candidates (lines 3-4) //validation to see if the candidates can form a pattern (group discovery)
3. adding new candidates (line 5) //final refinement (group pattern detection)

As mentioned in 4.3 there are various methods used to cluster moving objects or trajectories in the pre-processing stage in order to further mine the group patterns. This clustering stage is shared between other models of discussed algorithms including streaming models

Algorithm 1 Generic static approach

Input: $Tr, \delta_s, \delta_d/ \delta_g/ \delta_{dn}/ \delta_f/ \delta_r/ \delta_{ds}$ $\triangleright$ trajectory database data and a selection of thresholds

Output: all closed groups of moving objects

```

1: for <each time-slot, sub-trajectory/trajectory> do
2:   <cluster>
3: for <each potential candidate> do
4:   <extend current candidates list >
5:   if extended candidates match thresholds then
     return group of moving objects

```

as the key goal of this stage is the initial extraction of cluster-sets of objects at the given snapshots that have certain level of similarity whether it is a geometrical-based similarity such as distance, or density or other. Clustering can be either done on whole trajectories or sub-trajectories. The latter is further developed in a *partition-and-group* framework (Lee et al., 2007) which gives the advantage of discovering meaningful sub-trajectories which otherwise could have been missed. Majority of the key models in the static model group work with whole trajectories. However, while many algorithms from this model and other model groups use density-based clustering such as DBSCAN (e.g. convoy, swarm, gathering), in majority of cases there is no fixed clustering method in the pre-processing stage. The clustering methods in this first step can equally include k-means algorithm, or density-based ones as well as incremental clustering methods where clusters are computed incrementally from the last snapshot rather than from scratch at each snapshot (Li et al., 2010). The exception to these clustering methods are flock-based algorithms which instead of clustering, group the area of pattern search based on the fixed circular region which is expected to contain a minimum of moving objects within the given time. However, the outcome of this stage (lines 1-2) is similar as it results in a database of cluster-sets/predefined regions(flock-based models) at the given time-slot $C = \{C_1, C_2, \dots, C_n\}$. As defined in Section 4.1 this initial database of cluster-sets is used for candidates identification. The next stage (lines 3-4) is where the key differentiators between various models play a significant role.

The algorithms in this model group use all of the thresholds mentioned in Section 4.1. The extension of the candidates (lines 3-4) will be discussed from the perspective of both the thresholds used and the methods to create candidates from the database of cluster-sets extracted from the first pre-processing stage.

Swarm model is defined as a general type of moving object clusters containing at least minimum number of individuals forming the same cluster during at least certain number of timestamps. These timestamps are not consecutive which is differentiating this approach from

other ones in this group such as convoy and "classical" flock. If the group of moving objects is defined as O and the set of timestamps as T and if all objects in O are in the same cluster at any timestamp in T , then swarm is a pair (O, T) that satisfies the minimum constraints. However, due to this set up there is a high chance of extracting so-called redundant swarms which are a subset of the (O, T) pair if for instance the minimum number of individuals is set to 3 and the timestamps to 3 a group of moving objects $o1, o2, o3$ can fulfill these minimum constraints during timestamps $t1, t3, t9$ as it would during $t1, t3, t5, t9$. Therefore, the authors proposed the closed swarm concept where it is not required to output the subset $O' \subseteq O$ and $T' \subseteq T$ if the (O, T) is a swarm even in case if (O', T') fulfill the minimum requirements. Unfortunately, this proves to be a non-trivial task due to the size of all possible combinations. In the previously mentioned flock and convoy patterns the applied constraints including the consecutive timestamp significantly reduce the computation. In order to improve the search space and closure checking, the authors propose another new method called ObjectGrowth which uses two pruning rules: commonly used Apriori Pruning rule and a novel Backward Pruning rule which prevents the unnecessary further search. These two rules remove the unpromising candidates. In addition to this a Forward Closure Checking step is applied in order to report the closed swarm without using any space to store candidates, nor the extra processing time to perform this task.

Given the example, the minimum object number $\min_o$ is set up to 2 the same with the minimum number of timestamps $\min_t$. The nodes pruned by the Apriori rule are for instance $o1, o3, o2, o3, o3, o4$ and $o1, o2, o3$ due to the $Tmax(O) < \min_t$ in these cases. Even after pruning these object-sets there is still a high probability that the search space will be huge for instance in case of a significant number of objects that are in the same cluster during the whole time span. The example of the object-set $o1, o4$ shows that for any timestamp where these two are together also the $o2$ will be in the same cluster. Therefore, any other object-sets including $o1, o4$ but not including $o2$ cannot form a closed swarm so they can be pruned using the Backward Pruning rule. The example of object-set $o1, o2$ is not pruned by the two previous rules has a child node with $o1, o2, o4$ that matches the $Tmax(O)$. As per definition, the swarm $(O, Tmax(O))$ is closed if any superset O' of O with exactly one more object the $|Tmax(O')| < |Tmax(O)|$ which happened in this case.

In Convoy approach after the DBSCAN generates the set of clusters, they are compared to the existing candidates stored in V . If they share at least m common objects, then the existing candidates set is replaced by the intersection of common objects between a given cluster-set C and the candidate-set V and these objects are inserted into the V_{next} (that stores the current candidate list at a given time. The lifetime of the candidates is then increased until it reaches the required k -consecutive lifetime when it is reported as a convoy. These clusters

which do not fulfill the minimum common objects criterion with the existing candidates are identified as new candidates into V_{next} for a different convoy discovery. All candidates stored in V_{next} are then copied to V so that they are used for further processing in the next iteration (Jeung et al., 2010).

AGDP is based on the min distance and duration thresholds as well as the weight threshold which is the proportion of the time when the group of moving objects stays close so the higher the proportion the stronger and more meaningful the group discovery. Different from convoy where the existing candidates set is replaced by an intersection of common objects between a given cluster-set and the candidate-set, the AGP checks whether the candidates are close to each another at a given time-slot. The lengths of all valid segments are then accumulated in order to compute the weight of the candidate group. As a result the higher the weight, the better candidate set. The initial valid group pattern is a set of distinct moving objects and the *join* operation is used (from the classical Apriori algorithm) to generate the first candidate set. This candidate-set is then scanned to compute the weights to select the valid ones to be added to the next valid group pattern. Then again the *join* is used and the weight scanning is applied. The invalid subgroups are pruned in this process from the candidate set. Algorithm terminates when the thresholds of time duration is reached and calculates the union of the last stored valid group pattern with the previous ones (as the weight and distance/closeness are being checked at each iteration) (Peng LIM et al., 2004; Tsai et al., 2011; Wang et al., 2003, 2006)

In the flock approach where the clustering pre-processing is replaced by setting up a circular region (disk), the initial input data is firstly separated by equal time and equal motion direction resulting in a similar to cluster-set database used in previous models where the total number of cluster-sets is less or equal to the initial dataset. In order to find potential candidates (group discovery) Voronoi diagrams are used to check whether a subset of a size of at least m of the n - number of points is close together within the given circle of radius R . As a result these m points that are found within the smallest enclosing circle which match the radius threshold are considered the final group pattern. Leadership is a modification of the flock with an extra parameter requiring at least one member to be following the pattern for certain number of time-steps. This member is called a leader and the leader detection starts before the flock detection. The information about the circle containing a leader is stored. The algorithm terminates when both flock and the leader are found. Convergence takes the same parameters of a minimum group size and radius >0 and requires that at least m number of objects passes through the same circular region while encounter requires the minimum number of objects to be at the same time inside the same circular region of radius r . The convergence is found when a group of moving points can be found heading in the same

direction towards a point that can also be identified. For encounter a third dimension of time is added to mark the position of each object.

A moving cluster is combined of a sequence of continuous clusters which fulfill the criteria of a reaching a certain proportion of common members when intersecting the two neighbouring clusters (Kalnis et al., 2005). The threshold of overlapping (similarity) is being used to identify the moving clusters based on the percentage of common group members.

Evolving convoy follows same procedure as the regular convoy. However, different from convoy it allows members to leave and rejoin the group so the convoy may shrink or extend at certain times. This can be achieved by adding a gap/interval threshold.

5.2 Streaming data algorithms

Different from data stored in static databases, streaming data is not constant and may have a variable data distribution which changes in time as the data is streamed (Jiang and Gruenwald, 2006). Streaming trajectories are dynamic as the positions of objects are being continuously updated and therefore traditional ways of indexing such as R-tree or quad-tree require a high computational cost so new structures to store relevant data are required (Lee et al., 2003; Tang et al., 2012). In addition to this, the data from streaming trajectories may be subject to various delays and therefore the objects that potentially could be a part of the same group may not report their positions on time. When looking at data streams both offline streams and online streams can be distinguished (Jiang and Gruenwald, 2006). Offline streams are usually including regular bulk additions of data while online streams includes real-time updated data that arrive sequentially one after another. Different from static models, this group of streaming data algorithms is designed to work with both offline and online streaming data inputs which imposed different solutions to those presented in the aforementioned static examples. The static models require to load the whole trajectory of dataset into memory and therefore are not suitable for data streams which requires incremental processing. The algorithms in this streaming group include traveling companion and loose companion (Tang et al., 2012, 2013), loose traveling companion pattern and weakly continuous loose traveling companion pattern (Naserian et al., 2018), adaptive flock (Folino and Spezzano, 2002), online/incremental gathering and basic flock evaluation (Vieira et al., 2009). The generic streaming model is shown in Algorithm 2 and can be divided into the following steps:

1. initialize a temporary candidate set and cluster all objects from incoming stream/snapshot (lines 1 -3)

2. intersect existing candidates with temporary set and store the intersection result as new candidates (lines 4 -6)
3. validate new candidates (lines 7 -9)
4. return up-to-date group pattern to be processed in the next snapshot (line 10)

Algorithm 2 Generic streaming approach

Input: $T_s, \delta_s, V, \delta_d, \delta_g, \delta_{dn}, \delta_f, \delta_r, \delta_{ds}$ ▷ trajectory stream data, a selection of thresholds and candidates set

Output: O_{set} ▷ all up-to-date groups of moving objects

```

1: for <each snapshot> do
2:   <initialize a temporary candidate set>
3:   <cluster>
4: for <each potential candidate> do
5:   for <each cluster from  $T_s$ > do
6:     <new candidate = intersection of existing candidates with
       temporary candidates >
7:     if new candidate matches minimum thresholds then
8:       <add new candidate to temporary candidate set>
9:       if temporary candidate set matches thresholds then
10:        <add new candidates to existing candidates>
       return up-to-date group of moving objects ▷ to be processed in the next snapshots

```

Clustering at each snapshot of streaming data comes at high computational costs regardless of the clustering method chosen. Since the streamed objects positions are dynamic, maintaining spatial indexes at each snapshot such as presented in R-tree or Quadtree (Gudmundsson and van Kreveld, 2006; Lee et al., 2003) is not a scalable solution and therefore a new approach is required to handle streamed trajectories. These solutions include introducing new data structures to manage and store the temporary candidates and also to remove the obsolete records. These structures are based on various frameworks such as *clustering and intersection*, *filter-refinement* or online micro-clustering.

Tang et al following the clustering and intersection framework proposes a new structure called *traveling buddy* to maintain and store relationships between the objects in order to facilitate candidates discovery (Tang et al., 2012). While the objects change their positions in streaming trajectories, their relationships are less dynamic and can possibly stay constant over certain number of snapshots. The *buddy* structure replaces the spatial indexing and although its computational cost is $O(n^2)$ in the initial stage this step, it is performed only once resulting in creating a traveling buddy set which is much smaller than candidate cluster-set

and therefore requires less maintenance. The generic algorithm as presented in Algorithm 2 is then modified to perform a *buddy*-based intersection rather than cluster intersection based on existing candidates and temporary candidates. The *buddy* index consists of 3 elements *BID*, *ObjSet*, *CanID*, where *BID* is the *buddy*'s ID, *ObjSet* is the object members of the *buddy* structure and *CanID* records the IDs of candidates containing *buddy*. Therefore, the *buddy* structure does not contain the whole trajectory data about the object but only the relationships making it more lightweight to process.

Another approach is proposed by Mao et al who use the online micro-clustering and the sliding-window model to introduce a novel data structure called EF which can capture a change in trajectory cluster distribution. In order to cluster data streams, the authors adopted a *Micro-and-Macro clustering framework*, which was introduced by Aggarwal et al (Aggarwal et al., 2003) and followed by Lee and Li (Lee et al., 2007; Li et al., 2010). Similarly, to Jensen et al, who introduced a new summary data structure called *clustering feature* which reflects the key properties of a moving cluster for moving object cluster (Jensen et al., 2007), Mao et al. introduced their *EF* structure (Mao et al., 2016). The *EF* is in a form of summarization structure to store each new trajectory line segment that arrives at each time stamp. Eventually, they propose an offline macro-clustering which uses the *EF* structure to find the dynamic groups patterns. This approach is focused on in-cluster maintenance allowing to remove the obsolete records to speed up the overall processing. In order to remove obsolete records a *sliding-window* model is used which only takes the most recent data in the stream. *Sliding-window* is considered as a thresholds of size N and given the current time. Clustering is done on the data arriving in the interval window of the given window threshold and time.

Incremental gathering provides another solution of processing streaming data with an on-line crowd detection algorithm. The algorithm does not use a spatial index which comes with a high computational cost. Instead, it applies an on-line spatial join for pruning the clusters that are not relevant and uses the *traveling buddies* inside the clusters which should improve the processing time. This group pattern algorithm is the combination of gathering algorithm from static approach (Zheng et al., 2014) with a *traveling buddy* introduced in traveling companion (Tang et al., 2013) that is designed for data streams.

Different from the aforementioned approaches, Vieira in BFE uses the flock concept of fixed radius disks for processing streaming data by generating the candidate disks in each instance of time starting with the first instance and moving one instance at a time. This way every disk is evaluated and checked only against previous time instance and if they fulfill the criteria of candidates, they are kept for further processing while all others are removed. The

group pattern is reported as soon as the minimal thresholds are fulfilled so it is a relatively lightweight method without storing obsolete records.

5.3 Distributed/parallel algorithms

Continuously growing spatio-temporal data requires a big data processing approach. While the static models use memory processing which is not suitable for big trajectories, the streamed models propose a different approach to mine the streaming data but all of the algorithms in this group use relatively small datasets for processing. Distributed/parallel models for group pattern mining in this research area are relatively very few. The main algorithms in this group are distributed convoy (Orakzai et al., 2016), parallel adaptive flock (Folino and Spezzano, 2002), or another parallel approach using Swarm Intelligence - SPARROW (Jacob and Idicula, 2012). Another way of handling group pattern mining from large datasets are grid-based models such as moving flock (Andres Oswaldo Calderon Romero, 2011). A generic model for distributed algorithms is shown in Algorithm 3 and it can be divided into three key stages:

1. partitioning
2. local group pattern mining
3. merging results into a global result

Algorithm 3 Generic distributed approach

Input: partitions of Tr , δ_s , δ_d / δ_g / δ_{dn} / δ_f / δ_r / δ_{ds} $\triangleright$ partitions of trajectory data and a selection of thresholds

Output: O_{set} $\triangleright$ all groups of moving objects $\triangleright$ global result from all partitions

```

1: for <each moving object property/spatial/temporal dimension> do
2:   <partition data>  $\triangleright$  results in disjoint subsets
3: for <each disjoint subset> do
4:   <perform a local group pattern mining >
   return global result of group of moving objects

```

The partition of data into nodes can be done on object properties, their spatial or temporal dimensions. Partition on object properties may lead to a situation when spatially connected objects end up in different nodes and it is a high computational expense to communicate and retrieve the information between the nodes (Götze and Hagedorn, 2017; Orakzai et al., 2016). Similarly, partition on spatial dimension may lead to issues such as partial or missing

clusters which are hard to be computed from the local to global node. Different from spatial and object properties' partitioning, in the temporal partitioning the object moves only in one forward direction and therefore, searching for candidates does not need to be done across all nodes but across the previous or the next ones from the current node/partition. The second stage of the distributed algorithm model is the local pattern mining. The challenge with this stage lies in situations when the group pattern is split over several partitions and for instance if the time threshold requires 6 consecutive time stamps for the group to be together and the potential groups stay less than this given threshold, they will not be discovered on the local level but could potentially form a group if all partitions were taken into consideration as the rest of the group may be on a different partition. In order to overcome this issue, the algorithm is ran on each of the partitions and generates a local candidate which is then merged with local groups discovered in other partitions. Since the partition is done on the temporal dimension, only the neighbouring partitions need to be checked. The set of local group patterns can be presented as a union of the results from neighbouring partitions. The possible left, right and left-right opened candidates are used to perform this union as they have a potential to be extended. The result of this union is then added to the global result as it cannot be further extended.

Although focused on spatial dimension, SPARROW is another approach to parallel group discovery which leads to a discovery of moving flock patterns (Jacob and Idicula, 2012). The key stages of this algorithm to group discovery include the initial grouping the trajectory datasets based on fixed time intervals and then implementing SPARROW to find the flocks at each time interval. SPARROW considers four types of agents which depend on the density of neighbourhood around the moving points. They are differentiated with colours depending on their importance to the pattern discovery. Two types of agents are the one that indicate the extreme situation of red for interesting regions and white for not interesting regions and the speed of these agents is set up to 0 while the inbetween agents continue to move in order to find density interesting regions. Their direction and speed are calculated at each interval. A very recent paper from Xia et al. presents a parallel approach to frequent pattern discoveries (Xia et al., 2018). While it is not directly related to the group pattern mining which is the subject of this paper, they implement a parallel computing framework using map reduce which could be used for group pattern discovery. Firstly, they integrate the huge number of small files stored in HDFS using Sequence Files method to integrate them into a transaction database. This transaction database is then divided into several sub-transaction databases and these are allocated to different nodes of Hadoop cluster. A parallel computing is then to generate local frequent itemsets which are then aggregated to global itemsets. Each of this steps involves MapReduce paradigm.

Chapter 6

Universal Distributed and Parameter-free (USDP) algorithm

Taking into consideration the classification criteria that this survey was based on (data type, data size and group and movement parameters/thresholds), the proposed conceptual model aims to address the following criteria in group pattern discovery from spatio-temporal trajectories:

- big data processing via distributed computing
- handling streaming data
- parameter-free approach (or parameter-light)
- universal group pattern model

Since there are not many group pattern mining algorithms in this research area that propose a distributed framework and the moving object data is continuously growing, a distributed model to extract group patterns is in high demand. Furthermore, historical and static data input is not enough to capture the patterns as there many real-live applications require an incremental pattern mining from streaming data. The streaming data algorithms are a growing trend in spatio-temporal data mining. However, there are still many unresolved issues such as handling the dynamically changing data where it is hard to find a stable, constant element that can be referred to, or stored and that is not computationally expensive to be temporarily stored or processed. Another challenge is related to error-prone data due to imprecision of physical devices so introducing a structure to capture these changes and improving data precision is also important. Currently available models introduce a structure to store in memory certain information for future processing. However, since the streaming

data is potentially endless and storing and processing these temporary structures can be expensive from computational point of view, one of the possible solutions is to look at some form of universal data structure to capture the incoming data stream and use for clustering accordingly (Mao et al., 2016).

Parameter-free approach aims at reducing the bias coming from too many, or wrongly chosen parameters that can either cause the algorithm to fail, or to discover unwanted patterns (Keogh et al., 2004). This area of research is not commonly applied to group pattern mining from spatio-temporal trajectories with some exceptions such (Faghmous et al., 2013).

However, it could be used in the pre-processing stage to extract the widest possible cluster-sets of meaningful data for group pattern mining.

Lastly, a universal group pattern model that does not depend on specific pattern criteria, can be achieved by creating a customizable model that can take any thresholds defining groups. An interesting line of research which is not directly related to spatio-temporal data looks at thresholds estimation in order to machine-select best thresholds for the pattern mining but could potentially be used as an option to arbitrary human selection of the group defining thresholds (Mavridis et al., 2013).

Algorithm 4 Universal Streaming Distributed and Parameter-free approach

```

Input: Tr, Ts                                ▷ trajectory data, streaming and static
Output: Oset                                ▷ global result from all partitions, all groups of moving objects
1: if <input = Ts> then                                ▷ streaming data input
2:   <follow Algorithm 5>
3: else<continue>
4: for <each moving object's temporal dimension> do
5:   if <input = macro-cluster> then
6:     <partition by temporal dimension>
7:   else<continue>
8:   <partition data>                                ▷ results in disjoint subsets
9: for <each node, each partition> do
10:  <parameter-free clustering>                        ▷ results in local clusters
11: for <local cluster> do
12:  <perform a local group pattern mining >
13: for <each local candidates > do
14:  <merge closest candidates>                        ▷ checks local candidates with their closes
    candidates from closest nodes
15:  <update global candidates>
16: for <each global candidate> do
17:  <apply thresholds >                                ▷ checks if global result is a requested group
return global result of group of moving objects

```

Taking into consideration the above criteria, a framework of a Universal, Streaming, Distributed and Parameter-free (USDP) model/algorithm 4 is proposed to include the following key stages:

1. Distributed parameter-light pre-processing
2. Distributed streaming
3. Local and global pattern mining

The distributed pre-processing is based only on one criterion of partitioning data into distributed nodes based on temporal dimension which makes it parameter-light. Following current research in this matter, the partition based on time attribute rather than the spatial attribute has a benefit of a quick identification of possible candidates that could have been split into different nodes as the time dimension is uni-directional and moves forward rather than in many directions like the spatial attributes of the moving objects (Orakzai et al., 2016). In addition to this the streamed data will be easier to fit in the distributed model in order to check streamed candidates against the existing ones if we looked at the temporal dimension. Alternatively, a parameter-free incremental distributed hierarchical clustering can be considered which could free the initial partitioning of data from the temporal dimension criterion which has been assessed in the related literature (Jin et al., 2015).

Depending on the size of streaming data there are two approaches. If the streaming data is considered big, the first approach is to add the incremental streaming data to the main distributed partitioning and process it locally over partitioned nodes so that it follows the whole partitioning distributed model. If the streamed input data is manageable for initial pre-processing, then the second approach is to use the *Micro-and-Macro-Cluster* framework where the streaming input is firstly pre-processed using refinement techniques such as line segmentation and then micro and macro-clustering data in order to eventually cross-check it against the global result coming from the main distributed model.

To distributed streaming process is following:

1. For each new streamed input
2. do initial pre-processing (cluster input (line segmentation and other techniques to refine data if missing))
3. start: micro-clustering
4. For each cluster-set find the nearest micro-cluster matching the cluster-set

5. macro-cluster
6. return macro-cluster to be partitioned
7. partition macro-clusters to include in the nodes and process together with the existing data

Alternatively, a new data structure can be introduced to store candidates sets from the nodes that is lightweight enough to be kept for certain period of time and then cross-checked with the streamed candidates.

Lastly, the local and global pattern mining will be based on a set of group and movement defining thresholds which can be either user-defined, or machine-chosen.

Algorithm 5 USDP - Streaming

Input: Ts

▷ streaming data

Output: all valid macro-clusters

```

1: for <each new snapshot> do
2:   <line segmentation>
3:   <map-matching>
4: for <each line segment> do
5:   <find the nearest one within the sliding-window>
6:   <create micro-cluster>
7: for <each micro-cluster> do
8:   <check thresholds against other micro-clusters >
return all valid macro-clusters

```

Chapter 7

Conclusion

The focus of this paper was to provide an overview of the current research and its challenges in the area of group pattern discovery from spatio-temporal trajectories aiming at classifying the state-of-the-art algorithms. Despite of limitations resulting from the lack of access to full coding and tested datasets in majority of the cases, the available pseudo-codes allowed to classify algorithms. It has been achieved through creating a novel framework derived from traditional KDD data mining practice (Tan et al., 2014) which has been customized to handle big data processing as well as data streams. A classification of the existing group pattern algorithm has been based on this framework and following movement and group parameter criteria which extended the existing approaches of REMO (Laube and Imfeld, 2002) and movement patterns (Dodge et al., 2008). Key features of the analysed models have been extracted in the form of pseudo-codes and discussed based on the established classification criteria. With continuously growing spatio-temporal data there is a clear gap in group pattern discovery which does not scale up for big data and distributed computing. In addition to this many currently developed devices such as CCTV provide data streams while a significant majority of algorithms in spatio-temporal research area is designed to handle only static datasets. Due to the complexity of spatio-temporal data processing too many parameters leads to additional computational overhead and therefore there is a need for parameter-free or parameter-light models to optimise their performance. Taking all these key challenges into consideration, the author has proposed a theoretical model of Universal, Streaming, Distributed and Parameter-free (USDP) algorithm to fill the gap in this area. The algorithm follows distributed big data processing, is suitable for data streams and includes only one parameter in the pre-processing stage. Seeing that there are many thresholds differentiating the group patterns in this research area, the presented concept aims at being universal where group thresholds can be either user or machine-defined and can include a range of various thresholds depending on the application requirements. The proposed model

is in its early stage and requires further research, implementation and evaluation. The future implementation is considered via Spark platform as it is an efficient alternative to MapReduce via Hadoop which is currently being used for the distributed models in this research area. Spark includes GeoSpark and SpatialSpark which have spatial processing support inbuilt, however there is a gap in spatio-temporal tools available with some very few new options proposed such as STARK that complement the current tools with both spatial and temporal elements for big data processing (Götze and Hagedorn, 2017).

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